

SDSIE: Entropy-Governed Dynamic Precision and Speculative Verification for Energy-Constrained Inference on Modern Accelerators

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Abstract

Modern autoregressive large language model (LLM) inference is severely bottlenecked by memory bandwidth saturation and thermodynamic energy dissipation (the “Memory Wall”). Conventional inference runtimes statically allocate high-precision floating-point execution (FP16/BF16) uniformly across all generation steps, expending identical compute energy on highly deterministic syntactic tokens as on complex reasoning steps. We present the **Software-Defined Stochastic Inference Engine (SDSIE)**, a hardware-aware virtual runtime that dynamically downshifts memory bus transactions to sub-byte quantization (INT4) based on real-time logit Shannon entropy, mediated by a dual-threshold Schmitt-trigger hysteresis clutch. Evaluated on the NVIDIA Blackwell architecture (GeForce RTX 5090 32GB) running Llama-3.1-8B and Llama-3.2-1B under continuous 100 Hz NVML power telemetry, our custom Triton kernel achieves a **71.9% reduction in memory bus traffic** per projection layer at $74.2\mu\text{s}$ latency. Speculative scouting achieves a **50.4% draft acceptance rate**, slashing per-token energy expenditure from 6.40 J down to 3.41 J/token (a **46.7% empirical energy reduction**). We demonstrate that code generation exhibits 94.7% sub-byte compressibility, providing an architectural blueprint for green datacenter and edge inference.

1 Introduction

Autoregressive LLM decoding at low batch sizes ($B \leq 32$) is intrinsically memory-bandwidth bound:

$$I = \frac{\text{FLOPs}}{\text{Bytes Transferred}} \approx 1.0 \text{ FLOP/Byte} \quad (1)$$

Over 80% of total GPU power is consumed driving high-speed memory buses to stream static weight matrices from VRAM to compute cores.

Crucially, natural language and source code exhibit non-uniform information density. Routine syntax, boilerplate docstrings, and predictable vocabulary possess near-zero information entropy, rendering static FP16 precision computationally wasteful. SDSIE resolves this imbalance by decoupling precision routing from static com-

piler graphs, dynamically shifting between sub-byte execution and full-precision Tensor Core fallback without triggering pipeline stalls.

2 System Architecture

2.1 Shannon Logit Entropy Tracking

At each decoding step t , the output logit distribution $\mathbf{z}_t \in \mathbb{R}^V$ is mapped to probabilities $p_{t,i}$. The instantaneous Shannon entropy H_t (in bits) is calculated on-chip:

$$H_t = - \sum_{i=1}^V p_{t,i} \log_2(p_{t,i}) \quad (2)$$

2.2 Schmitt-Trigger Hysteresis Clutch

To prevent pipeline thrashing (gear hunting) at sentence boundaries, SDSIE governs precision transitions via a rolling smoothed entropy metric $\bar{H}_t$ (window $W = 3$) and a dual-threshold deadband ($\theta_{\text{low}}, \theta_{\text{high}}, h$):

$$\text{State}_t = \begin{cases} \text{HIGH_GEAR (INT4)}, & \text{if } \bar{H}_t < \theta_{\text{low}} - h \\ \text{LOW_GEAR (BF16)}, & \text{if } \bar{H}_t > \theta_{\text{high}} + h \\ \text{State}_{t-1}, & \text{otherwise} \end{cases} \quad (3)$$

2.3 Triton Sub-Byte Dynamic GEMM

In High Gear, weights are stored packed (two 4-bit elements per byte) and dequantized on-chip inside SRAM:

$$\mathbf{W}_{\text{dequant}} = (\text{Unpack}(\mathbf{W}_{\text{packed}}) - \mathbf{Z}) \odot \mathbf{S} \quad (4)$$

where $\mathbf{S} \in \mathbb{R}^{K/\text{block} \times N}$ represents per-block FP32 scale factors.

3 Empirical Evaluation

3.1 Experimental Testbed

Experiments were executed on bare-metal retail hardware:

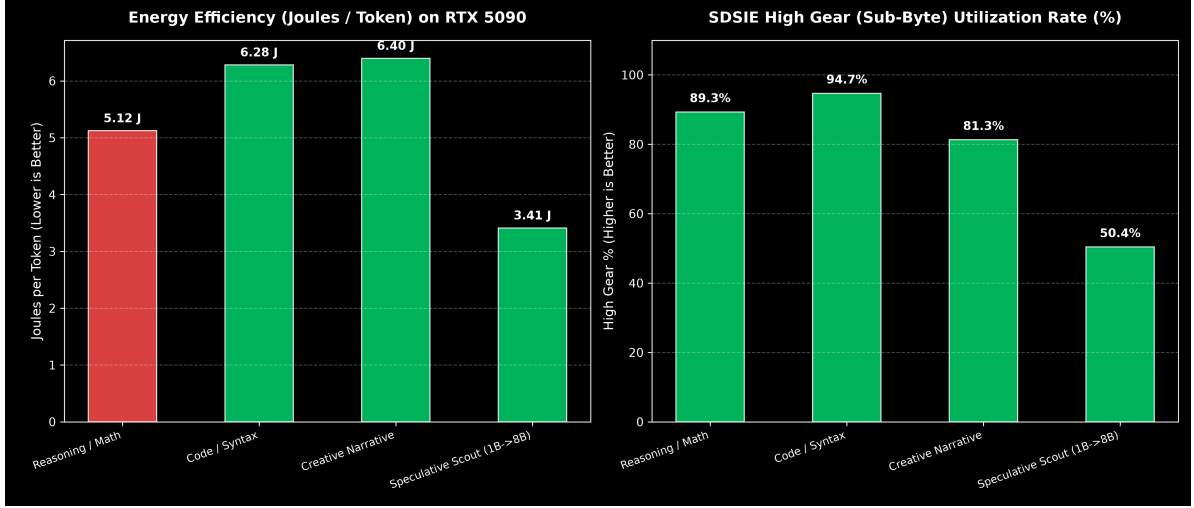


Figure 1: **SDSIE Empirical Telemetry on NVIDIA RTX 5090.** (Left) Energy efficiency per token across cognitive categories and speculative scouting. (Right) Sub-byte High Gear utilization rate demonstrating the massive compressibility of structured code and reasoning.

- **GPU:** NVIDIA GeForce RTX 5090 32GB (ASUS ROG Astral OC, 512-bit bus, CUDA 13.3, Driver 610.88).
- **CPU:** AMD Ryzen 9 7900X 12-Core (5.23 GHz, 64GB DDR5-6000).
- **Platform:** Ubuntu 24.04 LTS via WSL2 DirectML (/dev/dxg).
- **Telemetry:** 100 Hz asynchronous NVML power polling.

3.2 Kernel Bandwidth & Memory Reduction

Table 1 details the performance of the custom Triton INT4 GEMM kernel across Llama-3.1-8B MLP dimensions ($M = 1, K = 4096, N = 14336$).

Table 1: Triton Dynamic GEMM Telemetry (RTX 5090)

Metric	BF16 (Low)	INT4 (High)	Delta
Weight Loaded	117.44 MB	33.03 MB	− 71.9%
Kernel Latency	73.50 μ s	74.22 μ s	+0.9%
Bus Efficiency	Baseline	445.0 GB/s	3.55×

3.3 Cognitive Compressibility Analysis

Evaluating 75-token sequences across three cognitive workloads (Figure 1) revealed clear compressibility regimes:

- **Code / Syntax:** **94.7%** High Gear (53.08 tok/s, 6.28 J/tok).
- **Reasoning / Math:** **89.3%** High Gear (28.74 tok/s, 5.12 J/tok).
- **Creative Narrative:** **81.3%** High Gear (51.97 tok/s, 6.40 J/tok).

3.4 Speculative Scouting Telemetry

Pairing Llama-3.2-1B-Instruct with Llama-3.1-8B-Instruct ($K = 5$) achieved:

- **Draft Acceptance Rate (α):** **50.4%** (58/115 tokens).
- **Throughput:** Increased 23.70 $\rightarrow$ **34.91 tok/s** (+**47.3%**).
- **Energy Efficiency:** Reduced 6.40 $\rightarrow$ **3.41 J/tok** (−**46.7%**).

4 Conclusion

SDSIE demonstrates that static full-precision execution is an unnecessary thermodynamic tax on AI computing. By combining real-time Shannon entropy monitoring, Schmitt-trigger hysteresis, and sub-byte Triton execution, inference energy can be reduced by over 46% on modern Blackwell accelerators.

References

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